

KUSH PATHAK

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Summary

Third-year B.Tech Computer Science student with a proven track record in rigorous **Data Structures and Algorithms** problem-solving using **Java**. Passionate Backend Engineer focused on architecting scalable microservices using **Go (Golang)**, **Docker**, and **PostgreSQL**. Adept at translating complex algorithmic solutions into efficient, cloud-native applications and optimizing system architecture for high-concurrency environments.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY, BHOPAL

2023 – 2027

*B.Tech - Computer Science and Engineering - **CGPA: 8.32***

Bhopal, Madhya Pradesh

TECHNICAL SKILLS

Languages: Java (Advanced), Go (Golang), SQL, HTML, CSS

Backend Technologies: RESTful APIs, gRPC, Microservices, ACID Transactions

Tools & Platforms: Docker, PostgreSQL, AWS (EC2, S3), Git, Postman

Coursework: Data Structures & Algorithms, OOPs, OS, DBMS, Computer Networks

ACHIEVEMENTS & PROBLEM SOLVING

- **Algorithmic Problem Solving:** Solved **250+** problems on LeetCode, exclusively using **Java** to implement highly optimized solutions.
- **Competitive Programming:** Maintain a LeetCode contest rating of **1500+** and actively participate in structured problem-solving assessments.

PROJECTS

FinTech Banking API 🏦 | Go, PostgreSQL, Docker, AWS

- Architecting a secure, high-performance banking ledger service using **Golang** and **PostgreSQL**.
- Implemented **ACID compliant transactions** with rigorous locking mechanisms to prevent race conditions.
- Optimized ledger querying and retrieval algorithms, effectively reducing transaction processing time complexity and improving overall API throughput.
- Designing dual-protocol APIs (REST & gRPC) secured with **PASETO** tokens for stateless authentication.
- Containerized the application using **Docker** and established CI/CD pipelines via GitHub Actions for automated testing.

Android Malware Detection System 📱 | Machine Learning, Cyber Security

- Engineered a dynamic emulation pipeline in **Android Studio** to securely execute applications, utilizing **PCAPdroid** to intercept and record live network traffic.
- Curated a custom dataset of 100 Android applications by developing **Python** and **Scapy** scripts to parse raw PCAP binaries into structured TCP/UDP flow metrics.
- Evaluated a dual-phase classification engine, achieving **95% accuracy in static analysis (k-NN)** and **75% accuracy in dynamic behavioral analysis (Logistic Regression)**.

CERTIFICATIONS

- **Introduction to Machine Learning** — NPTEL / IIT Madras
- **Marketing Analytics** — NPTEL / IIT Kharagpur
- **Front End Development** — IBM
- **The Bits and Bytes of Computer Networking** — Google